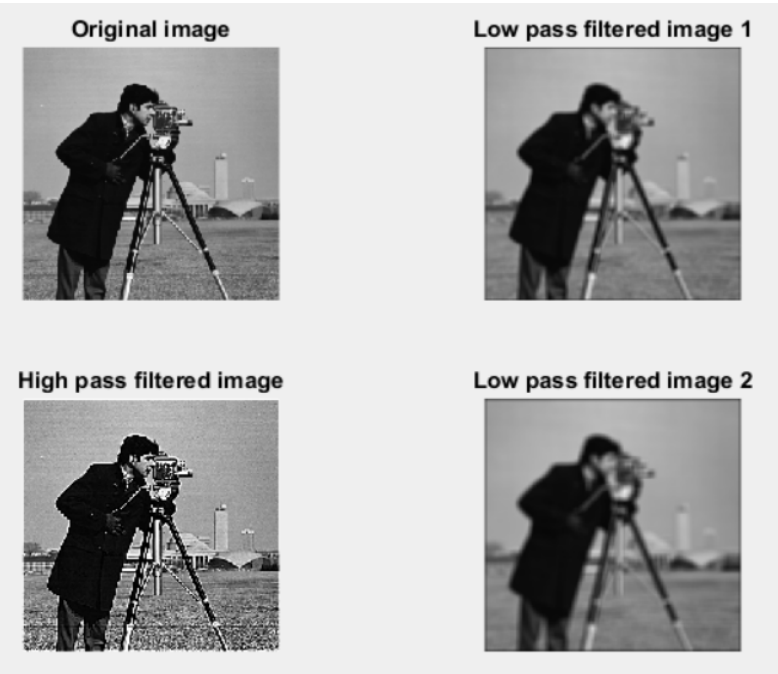
CYCLE 2 PROGRAMS

7.1 Medical Image smoothing (LPF & HPF filter)

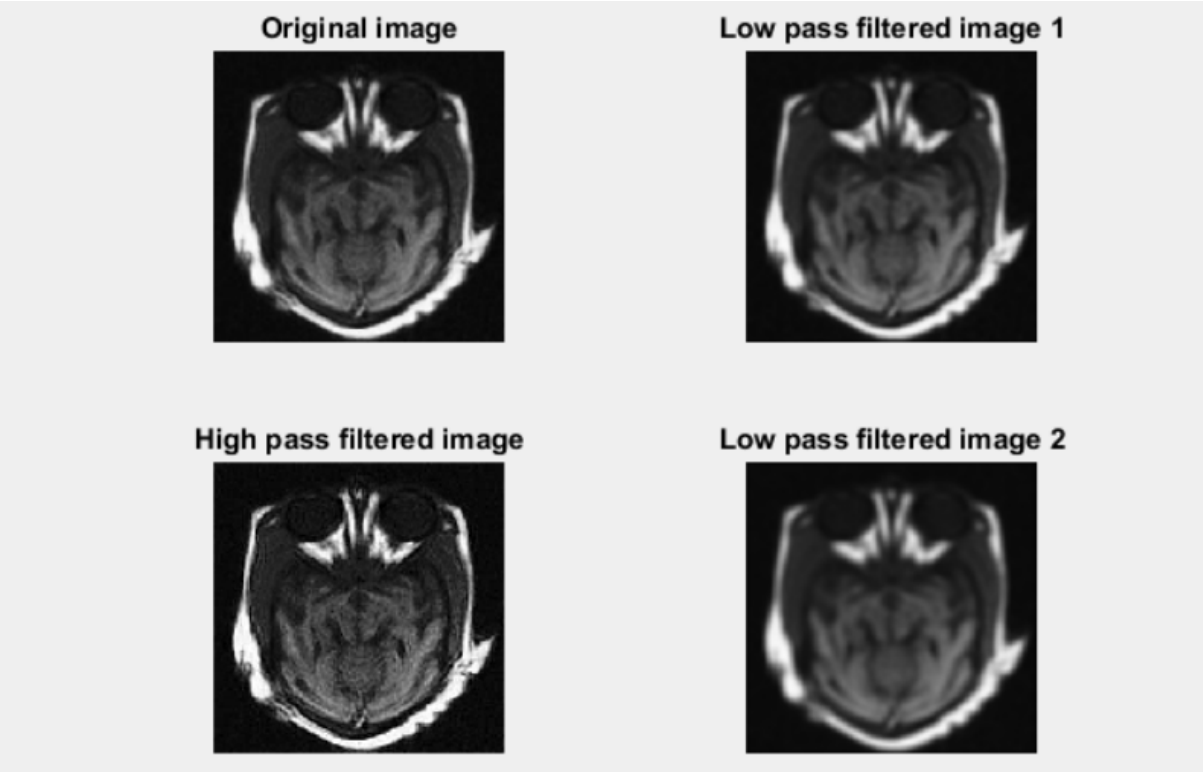
```
close all;
clear all;
clc;
a=imread('cameraman.tif');
low=fspecial('average',[5 5]);
low1=imfilter(a,low);
high=fspecial('unsharp');
high1=imfilter(a,high);
subplot(221);imshow(a);title('Original image');
subplot(222);imshow(low1);title('Low pass filtered image 1');
subplot(223);imshow(high1);title('High pass filtered image');
low=fspecial('average',[7 7]);
low2=imfilter(a,low);
subplot(224);imshow(low2);title('Low pass filtered image 2');
```

OUTPUT:



```
close all;
clear all;
clc;
a=imread('C:\Program Files\MATLAB\MATLAB Production
Server\R2015a\bin\MIPIMAGES\med256B.jpg');
low=fspecial('average',[5 5]);
low1=imfilter(a,low);
high=fspecial('unsharp');
high1=imfilter(a,high);
subplot(221);imshow(a);title('Original image');
subplot(222);imshow(low1);title('Low pass filtered image 1');
subplot(223);imshow(high1);title('High pass filtered image');
low=fspecial('average',[7 7]);
low2=imfilter(a,low);
subplot(224);imshow(low2);title('Low pass filtered image 2');
```

OUTPUT:



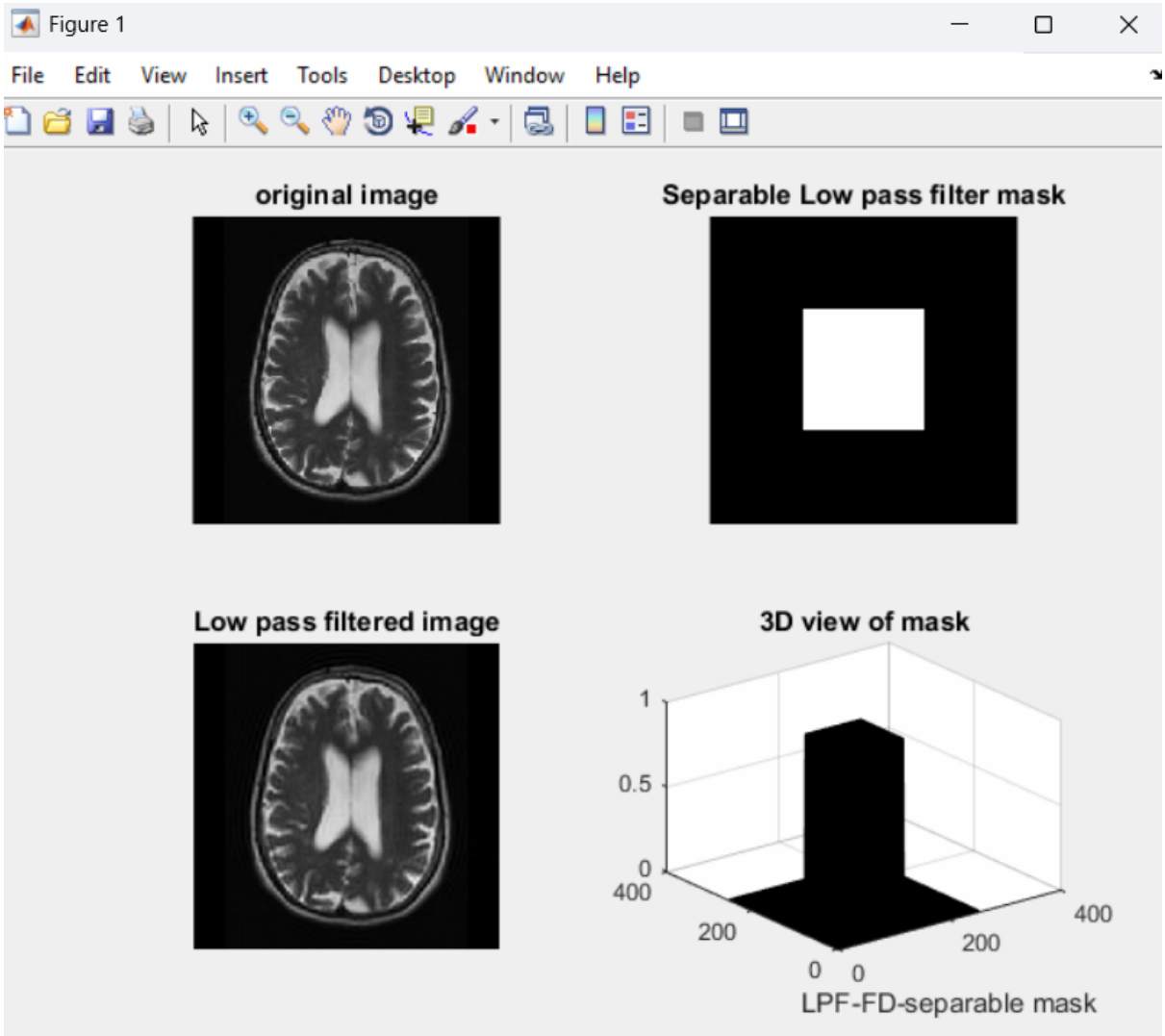
7.2 Image smoothing (LPF - FREQUENCY DOMAIN)

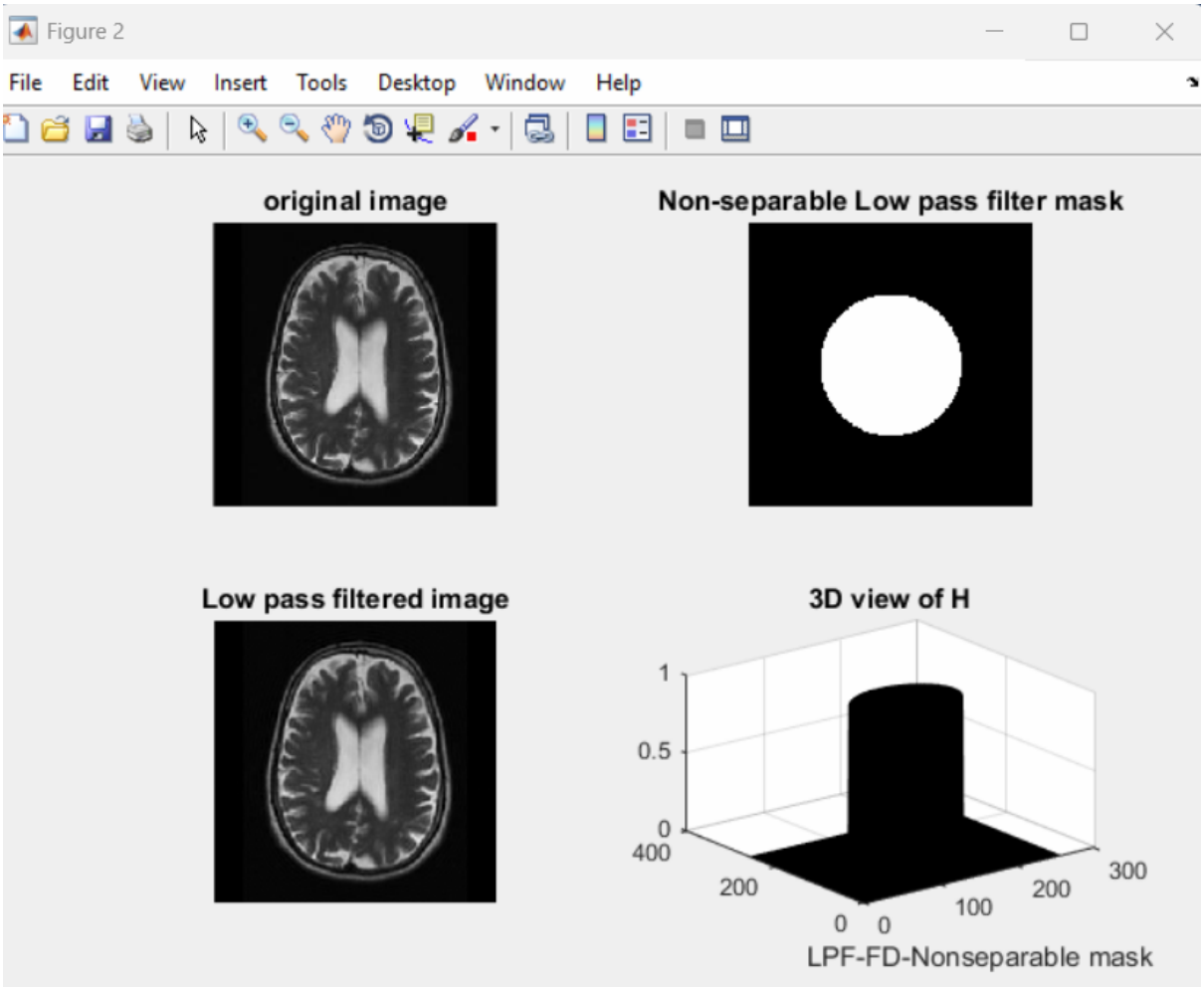
```
clc;
clear all;
close all;
a=im2double(imread('C:\Program Files\MATLAB\MATLAB Production
Server\R2015a\bin\MIPIMAGES\mri_brain.jpg')) ;
[m n]=size(a);
b=fft2(a);
c=fftshift(b);
mask=zeros(m,n);
p=100;
q=100;
for i=(m-p)/2:(m+p)/2
   for j=(n-q)/2:(n+q)/2
      mask(i,j)=1;
   end
end

%filtering
d1=c.*mask;
d2=fftshift(d1);
e=abs(ifft2(d2));
subplot(2,2,1),imshow(a),title('original image');
subplot(2,2,2),imshow(mask),title('Separable Low pass filter mask');
subplot(2,2,3),imshow(e),title('Low pass filtered image');
subplot(224),surf(mask),title('3D view of mask');
xlabel(sprintf('LPF-FD-separable mask'));
figure();
[u,v]=freqspace(256,'meshgrid');
H=zeros(256,256);
d=sqrt(u.^2+v.^2)<0.5;
H(d)=1;
x1=c.*H;
```

```
x2=fftshift(x1);
x3=abs(ifft2(x2));
subplot(2,2,1),imshow(a),title('original image');
subplot(2,2,2),imshow(H),title('Non-separable Low pass filter mask');
subplot(2,2,3),imshow(x3),title('Low pass filtered image');
subplot(224),surf(H),title('3D view of H');
xlabel(sprintf('LPF-FD-Nonseparable mask'));
```

OUTPUT:

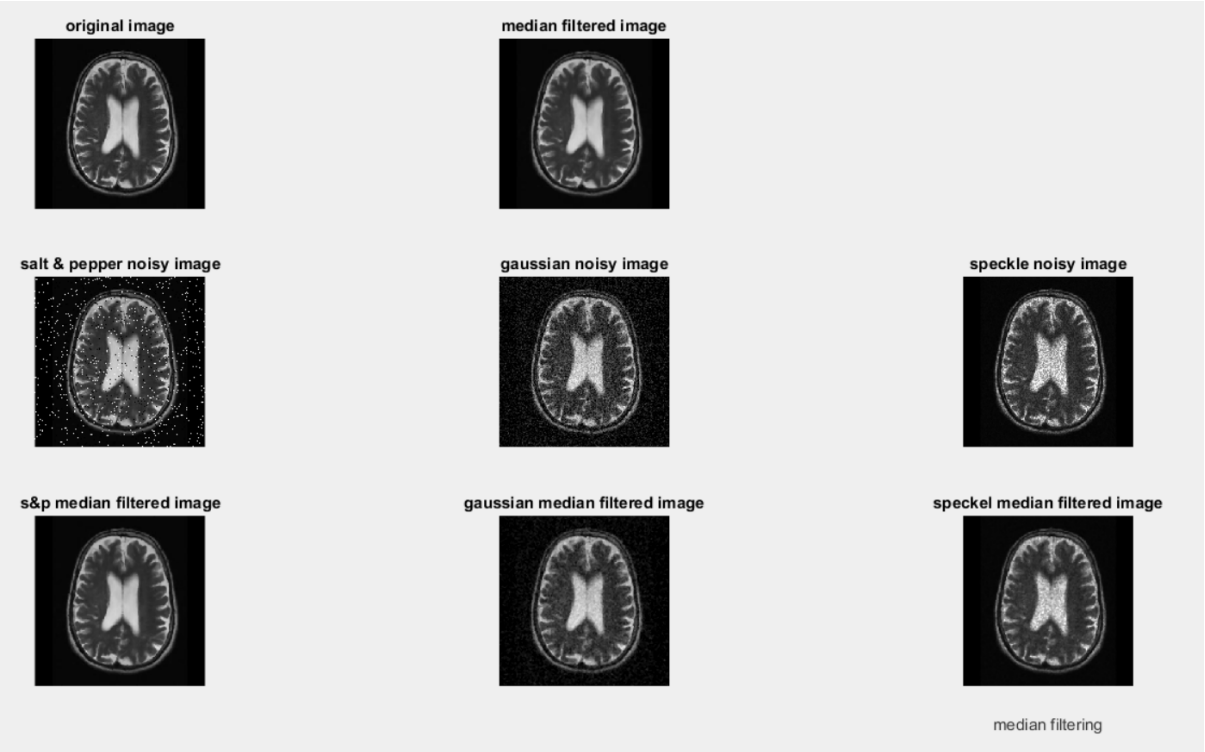



Figure 2
File Edit View Insert Tools Desktop Window Help
original image
Non-separable Low pass filter mask
Low pass filtered image
3D view of H
1
0.5
0
400
200
0
0
100
200
300
LPF-FD-Nonseparable mask

7.3 Medical Image smoothing (median filter)

```
% Medical Image smoothing (median filter)
 close all; clear all; clc;
f=imread(' C:\Program Files\MATLAB\MATLAB Production
Server\R2015a\bin\MIPIMAGES\mri_brain.jpg');
g=medfilt2(f,[3 3]);
  subplot(331),imshow(f);title('original image');
  subplot(332),imshow(uint8(g));title('median filtered image');
  n1=imnoise(f,'salt & pepper');
  subplot(334);
  imshow(n1);
  title('salt & pepper noisy image');
  n2=imnoise(f,'gaussian');
  subplot(335)
  imshow(n2)
  title('gaussian noisy image');
  n3=imnoise(f,'speckle');
  subplot(336)
  imshow(n3)
  title('speckle noisy image');
  h2=medfilt2(n1,[3 3]);
  subplot(337);
  imshow(h2);
  title('s&p median filtered image')
  h3=medfilt2(n2,[3 3]);
  subplot(338);
  imshow(h3);
  title('gaussian median filtered image');
  h4=medfilt2(n3,[3 3]);
  subplot(339);
  imshow(h4);
  title('speckel median filtered image');
  xlabel(sprintf('median filtering'));
```

OUTPUT:

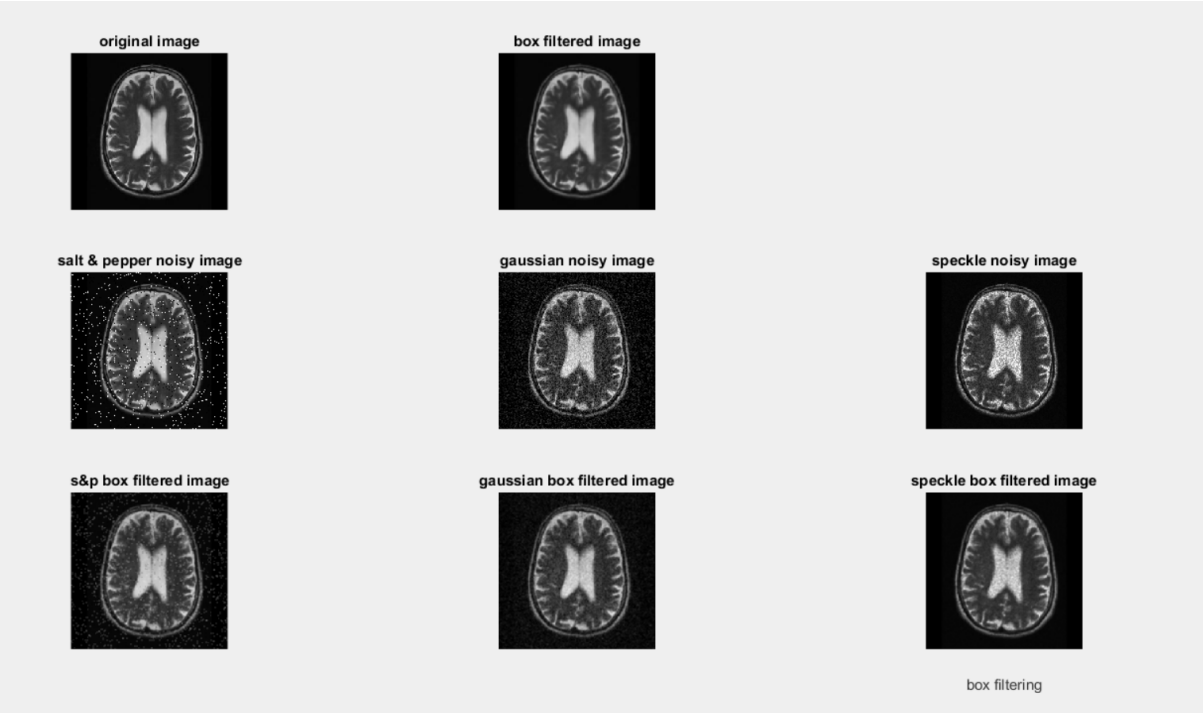

original image
median filtered image
salt & pepper noisy image
gaussian noisy image
speckle noisy image
s&p median filtered image
gaussian median filtered image
speckel median filtered image
median filtering

7.4 Medical Image smoothing (mean filter)

```
% Medical Image smoothing (mean filter)
close all;
clear all;
clc;
f=imread('C:\Program Files\MATLAB\MATLAB Production
Server\R2015a\bin\MIPIMAGES\mri_brain.jpg');
h=1/9*[1 1 1
       1 1 1
       1 1 1];
  g=conv2(f,h,'same');
  subplot(331),imshow(f);title('original image');
 subplot(332),imshow(uint8(g));title('box filtered image');
  n1=imnoise(f,'salt & pepper');
 subplot(334);
 imshow(n1);
 title('salt & pepper noisy image');
 n2=imnoise(f,'gaussian');
 subplot(335);
 imshow(n2);
 title('gaussian noisy image');
 n3=imnoise(f,'speckle');
 subplot(336);
 imshow(n3);
 title('speckle noisy image');
 g1=conv2(n1,h,'same');
 subplot(337);
 imshow(uint8(g1));
 title('s&p box filtered image');
 g2=conv2(n2,h,'same');
 subplot(338);
 imshow(uint8(g2));
 title('gaussian box filtered image');
```

```
  g3=conv2(n3,h,'same');
  subplot(339);
  imshow(uint8(g3));
  title('speckle box filtered image');
xlabel(sprintf('box filtering'));
```

OUTPUT:



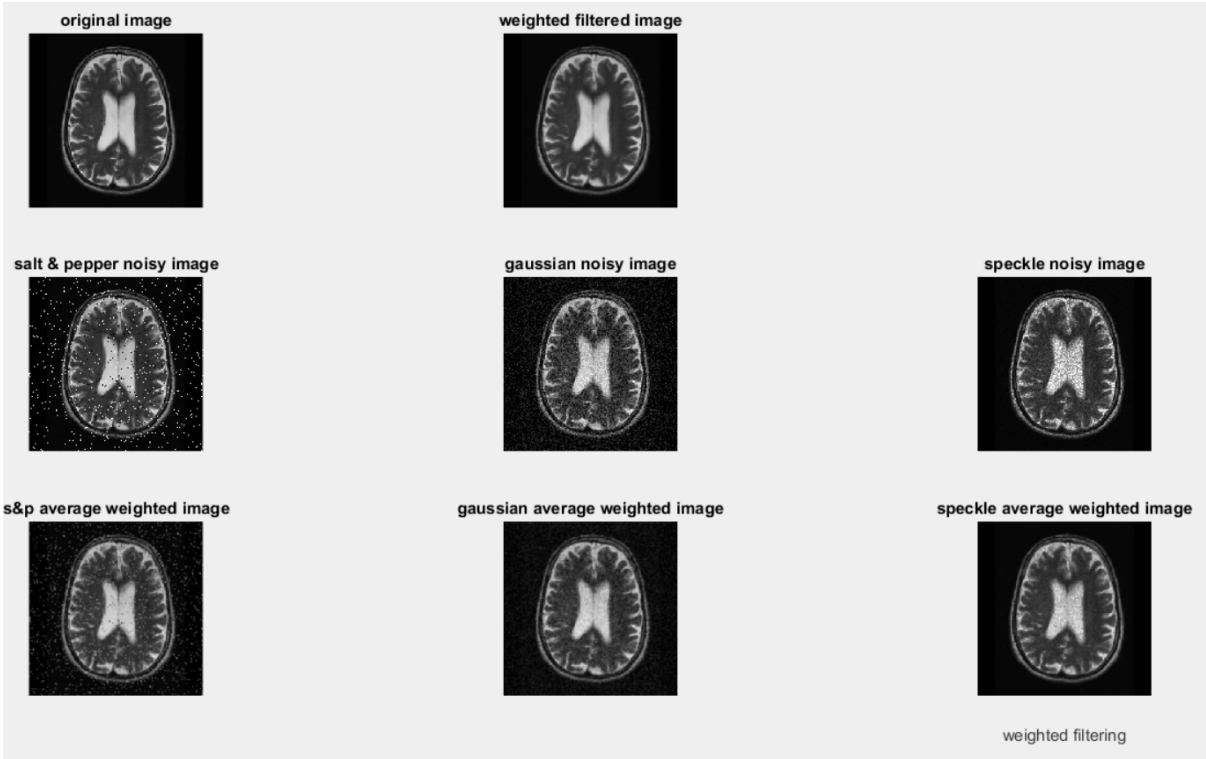
7.5 Medical Image smoothing (weighted average filter)

```
% Medical Image smoothing (weighted average filter)
close all;
clear all;
clc;
f=imread('C:\Program Files\MATLAB\MATLAB Production
Server\R2015a\bin\MIPIMAGES\mri_brain.jpg');
h=1/16*[1 2 1
        2 4 2
        1 2 1];
  g=conv2(f,h,'same');
     subplot(331),imshow(f);title('original image');
 subplot(332),imshow(uint8(g));title('weighted filtered image');
  n1=imnoise(f,'salt & pepper');
 subplot(334);
 imshow(n1);
 title('salt & pepper noisy image');
 n2=imnoise(f,'gaussian');
 subplot(335);
 imshow(n2);
 title('gaussian noisy image');
 n3=imnoise(f,'speckle');
 subplot(336);
 imshow(n3);
 title('speckle noisy image');
 g1=conv2(n1,h, 'same');
 subplot(337);
 imshow(uint8(g1));
 title('s&p average weighted image');
 g2=conv2(n2,h, 'same');
 subplot(338);
 imshow(uint8(g2));
 title('gaussian average weighted image');
```

```
g3=conv2(n3,h, 'same');
subplot(339);
imshow(uint8(g3));
title('speckle average weighted image');

xlabel(sprintf('weighted filtering'));
```

OUTPUT:



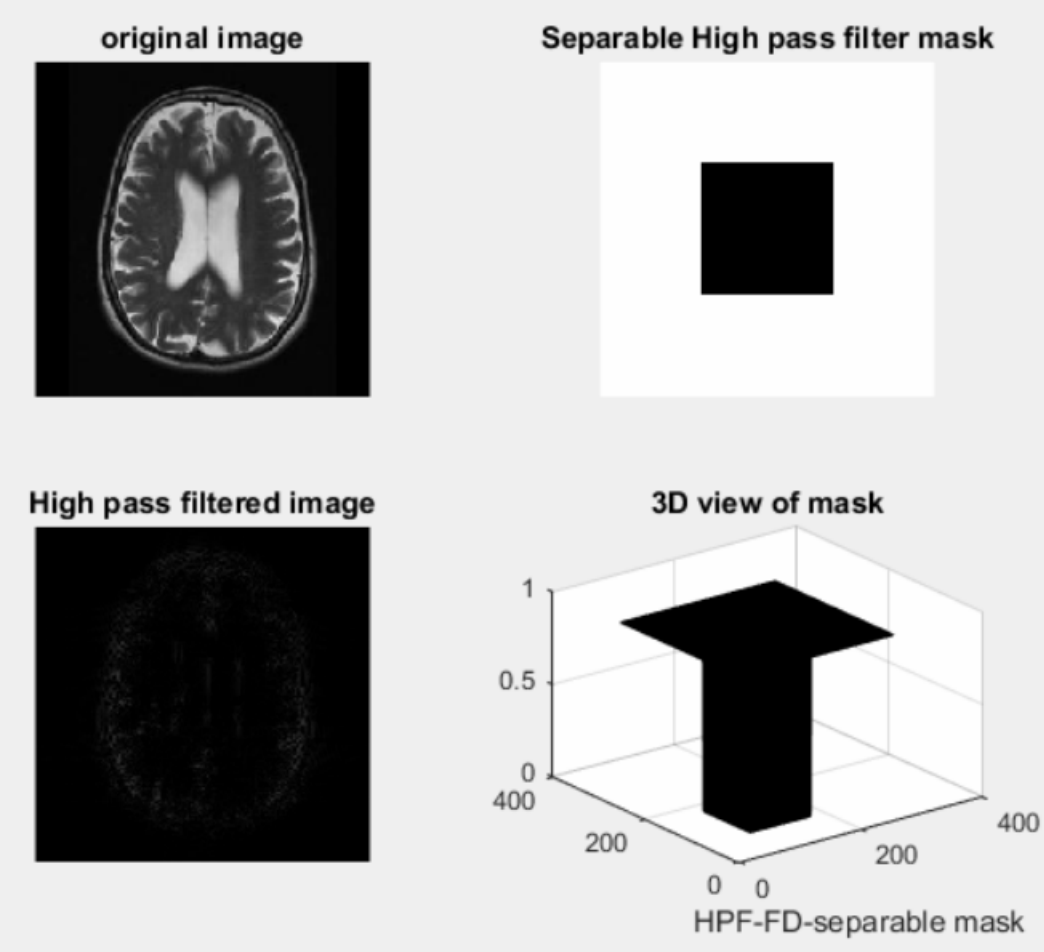
8.1 image Sharpening (HPF filter)

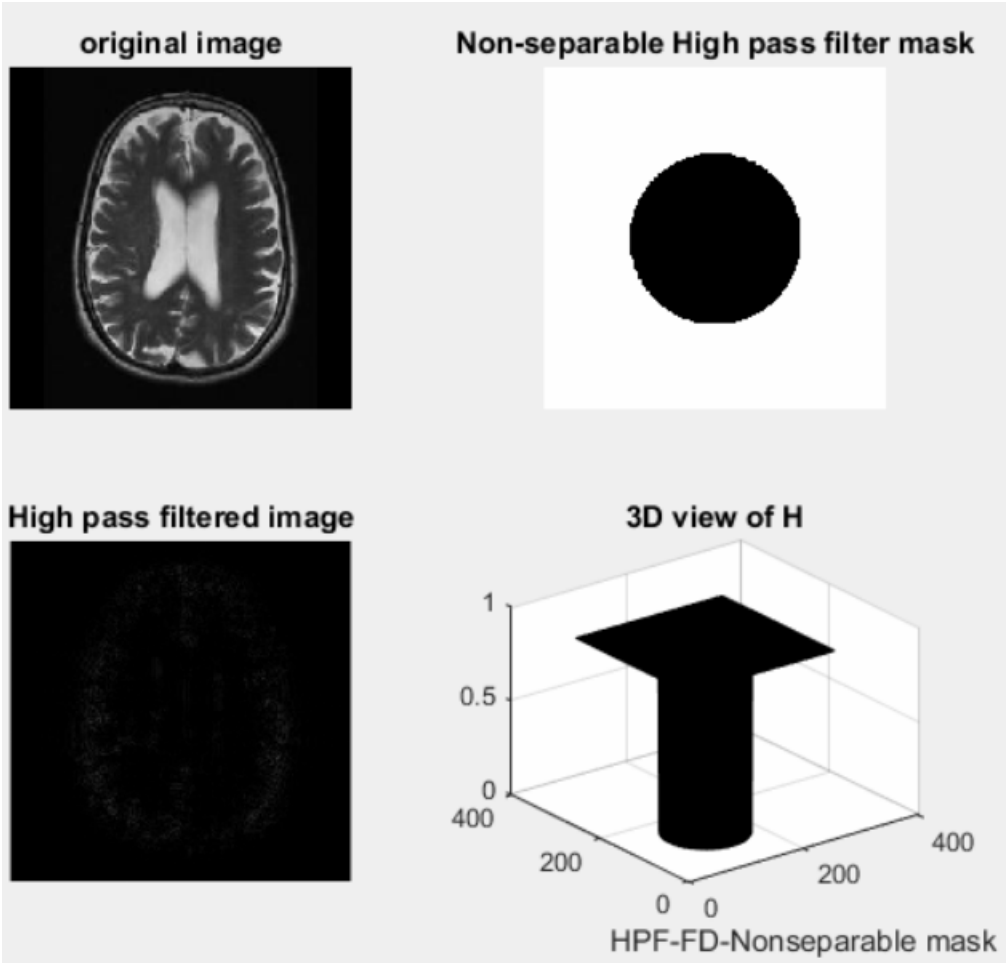
```
% Image sharpening (HPF filter)
clc;
clear all;
close all;
a=im2double(imread('C:\Program Files\MATLAB\MATLAB Production
Server\R2015a\bin\MIPIMAGES\mri_brain.jpg'));
[m n]=size(a);
b=fft2(a);
c=fftshift(b);
mask=ones(m,n);
p=100;
q=100;
for i=(m-p)/2:(m+p)/2
   for j=(n-q)/2:(n+q)/2
      mask(i,j)=0;
   end
end

%filtering
d1=c.*mask;
d2=fftshift(d1);
e=abs(ifft2(d2));
subplot(2,2,1),imshow(a),title('original image');
subplot(2,2,2),imshow(mask),title('Separable High pass filter mask');
subplot(2,2,3),imshow(e),title('High pass filtered image');
subplot(224),surf(mask),title('3D view of mask');
xlabel(sprintf('HPF-FD-separable mask'));
figure();
[u,v]=freqspace(256,'meshgrid');
H=ones(256,256);
d=sqrt(u.^2+v.^2)<0.5;
H(d)=0;
x1=c.*H;
```

```
x2=fftshift(x1);
x3=abs(ifft2(x2))
subplot(2,2,1),imshow(a),title('original image');
subplot(2,2,2),imshow(H),title('Non-separable High pass filter mask');
subplot(2,2,3),imshow(x3),title('High pass filtered image');
subplot(224),surf(H),title('3D view of H');
xlabel(sprintf('HPF-FD-Nonseparable mask'));
```

OUTPUT:

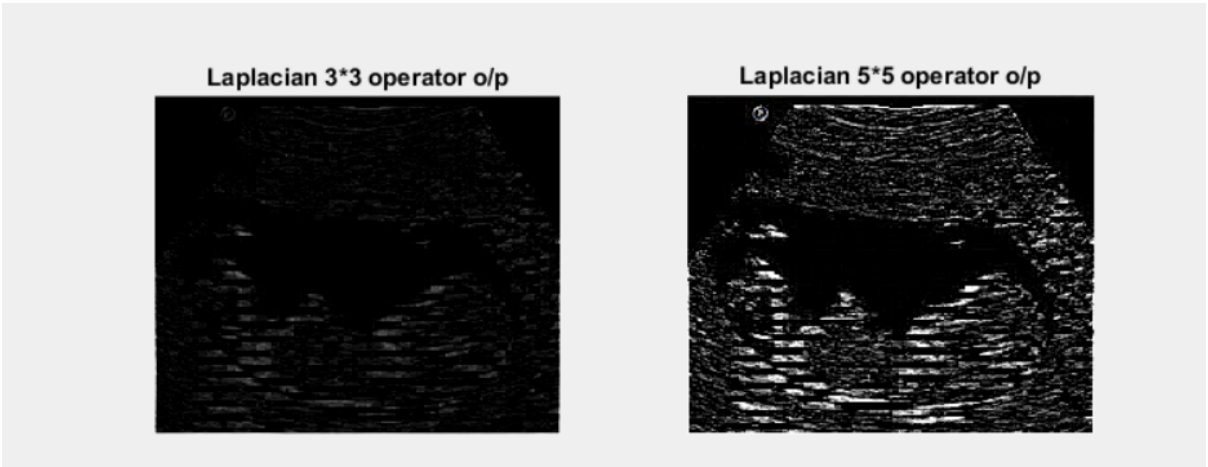



original image
Non-separable High pass filter mask
High pass filtered image
3D view of H
1
0.5
0
400
200
0
0
200
400
HPF-FD-Nonseparable mask

8.2 Medical Image Sharpening (Derivative filter)

```
%Medical Image sharpening (Derivative filter)
close all;
clear all;
clc;
a=imread('C:\Program Files\MATLAB\MATLAB Production
Server\R2015a\bin\MIPIMAGES\ultrasound.jpg');
imshow(a);title('Original image');
b1=[0 -1 0;-1 4 -1;0 -1 0];
b2=[0 0 -1 0 0;0 -1 -2 -1 0;-1 -2 16 -2 -1;0 -1 -2 -1 0;0 0 -1 0 0 ];
a1=imfilter(a,b1);
a2=imfilter(a,b2);
figure(2);
subplot(1,2,1);imshow(a1);title('Laplacian 3*3 operator o/p');
subplot(1,2,2);imshow(a2);title('Laplacian 5*5 operator o/p');
```


Figure 1
File Edit View Insert Tools Desktop Window Help
Original image

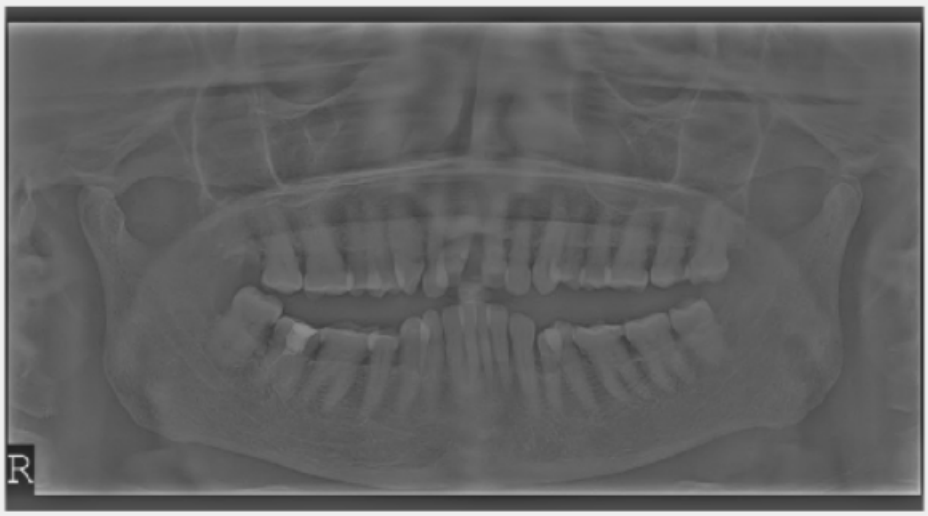

Laplacian 3*3 operator o/p
Laplacian 5*5 operator o/p

9. Implementation of homomorphic filtering technique for image enhancement

```
clear all;close all;clc;
r=imread('C:\Program Files\MATLAB\MATLAB Production
Server\R2015a\bin\MIPIMAGES\dental.jpg');
a=rgb2gray(r);
b=im2double(a);
I=log(1+b);
[m,n]=size(I);
%% centering
for i=1:m
   for j=1:n
      p(i,j)=((-1)^(i+j))*I(i,j);
   end
end
bf=fft2(p);
%% Design of H(u,v)
for u=1:m
   for v=1:n
      d=((u-m/2)^2+(v-n/2)^2)^(1/2);
%    if d<30
%       H(u,v)=1;
%    else
%       H(u,v)=0;
%    end
H(u,v)=1-exp(-d^2/100);%Gaussian HPF
   end
end
G=bf.*H;
g=real(ifft2(G));
%% De-center
for i=1:m
```

```
    for j=1:n
        g1(i,j)=((-1)^(i+j))*g(i,j);
    end
end
g2=(exp(g1)-1);
subplot(211);imshow(b);
subplot(212);imshow(g2,[]);
```





10. Error measures using MSE and NMSE

```
%Error measures using MSE and NMSE

clc;       % Clear the command window.
close all;  % Close all figures (except those of imtool.)
clear;      % Erase all existing variables.
I = imread(' C:\Program Files\MATLAB\MATLAB Production
Server\R2015a\bin\MIPIMAGES\mri_brain.jpg');
[rows columns] = size(I);
% addition of graininess (i.e. noise)
I_noise = imnoise(I, 'gaussian', 0.09);
% the average of 3^2, or 9 values(filters the multidimensional array A with the
multidimensional filter h)
 h = ones(3,3) / 3^2;
 I_filt = imfilter(I_noise,h);
I_1=double(I(:));
I_noise1=double(I_noise(:));
I_filt1=double(I_filt(:));

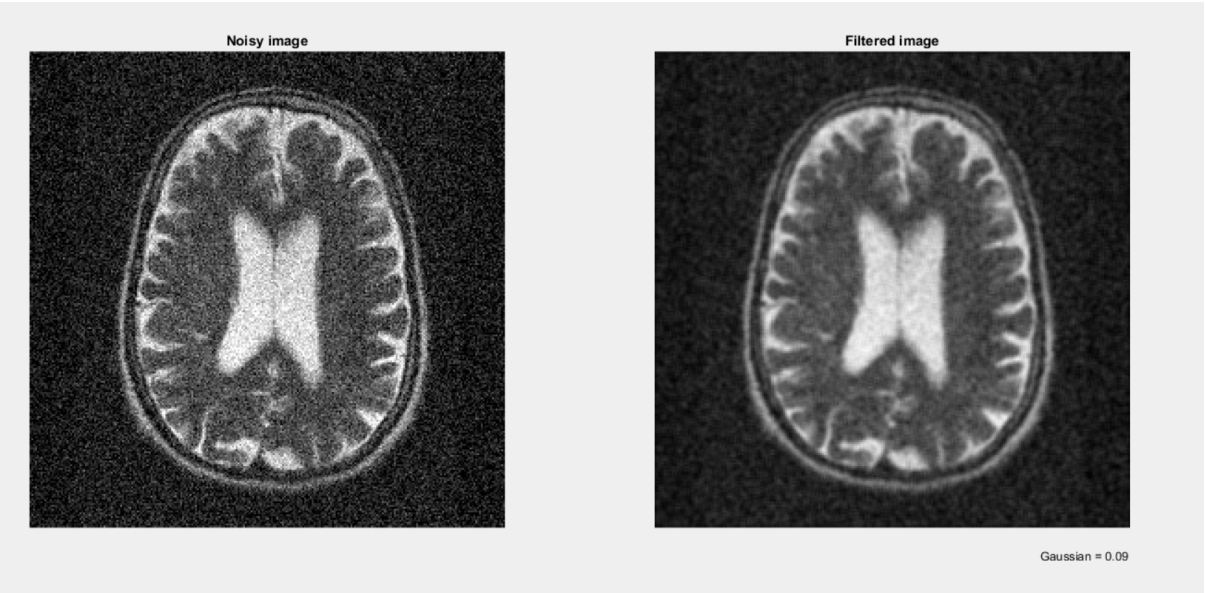
% Mean Square Error
MSE = sum(sum((I_filt1-I_1) .^ 2)) / (rows * columns);

%Normalized Mean Square
NMSE = sum(sum((I_filt1-I_1) .^ 2))/ sum(sum((I_noise1-I_1).^2));
 fprintf('\n The MSE value is  %0.4f %0.4f \n', MSE);
fprintf('\n The NMSE value is  %0.4f %0.4f \n', NMSE);
  %Plot original & filtered figure
subplot(1,2,1), imshow(I_noise), title('Noisy image')
subplot(1,2,2), imshow(I_filt), title('Filtered image')
text(size(I,2),size(I,1)+15, ...
     'Gaussian = 0.09', ...
     'FontSize',10,'HorizontalAlignment','right');
```

OUTPUT:

The MSE value is 670.0258

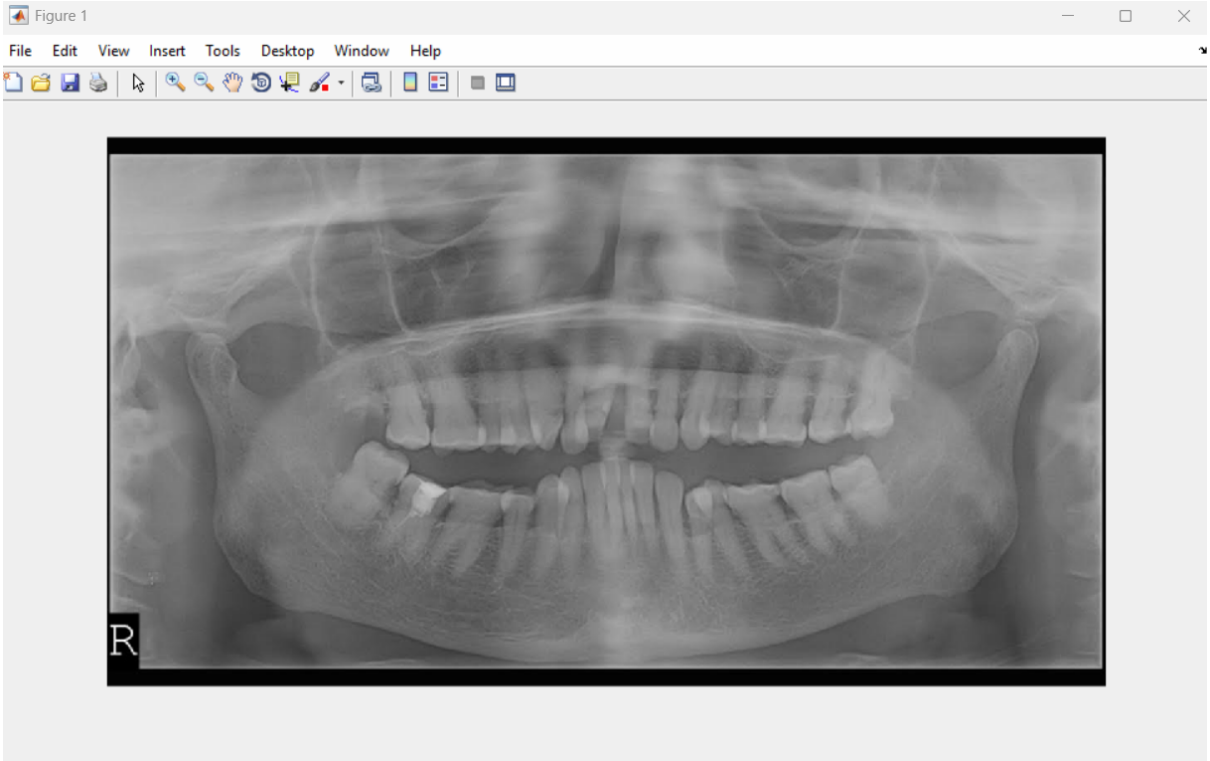
The NMSE value is 0.5892

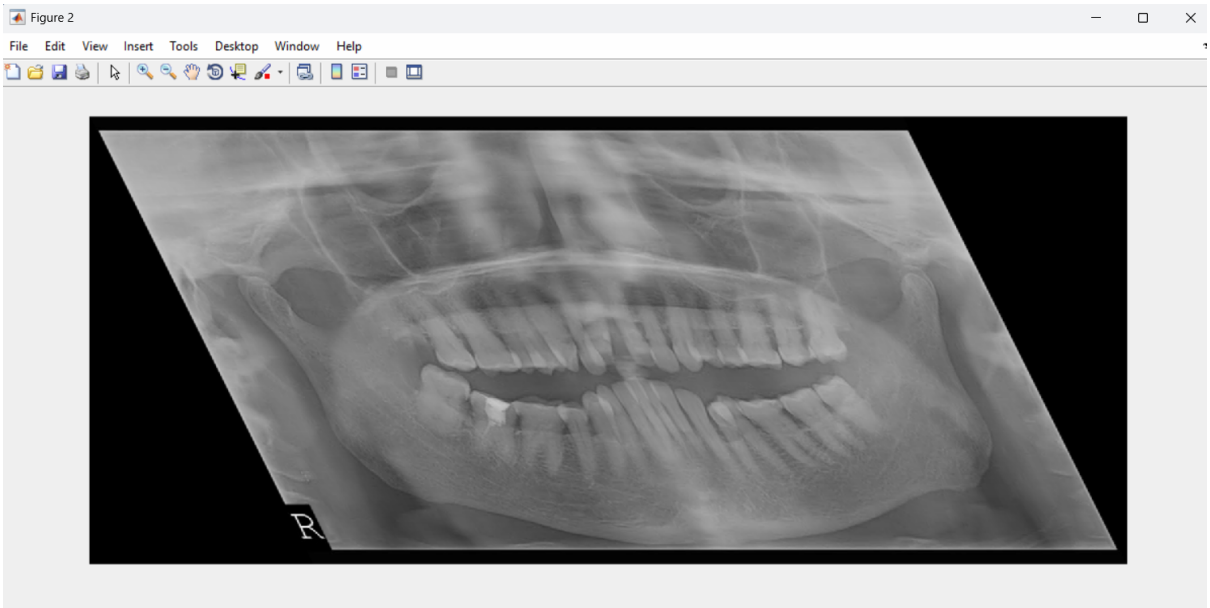


11.1 Geometric transformation -2D

```
clear all;close all;clc;
%Edge detection using sobel mask
r=imread('C:\Program Files\MATLAB\MATLAB Production Server\R2015a\bin\MIPIMAGES\dental.jpg');
I=rgb2gray(r);
imshow(I)
tform = affine2d([1 0 0; .5 1 0; 0 0 1])
J = imwarp(I,tform);
figure
imshow(J)
```

OUTPUT:



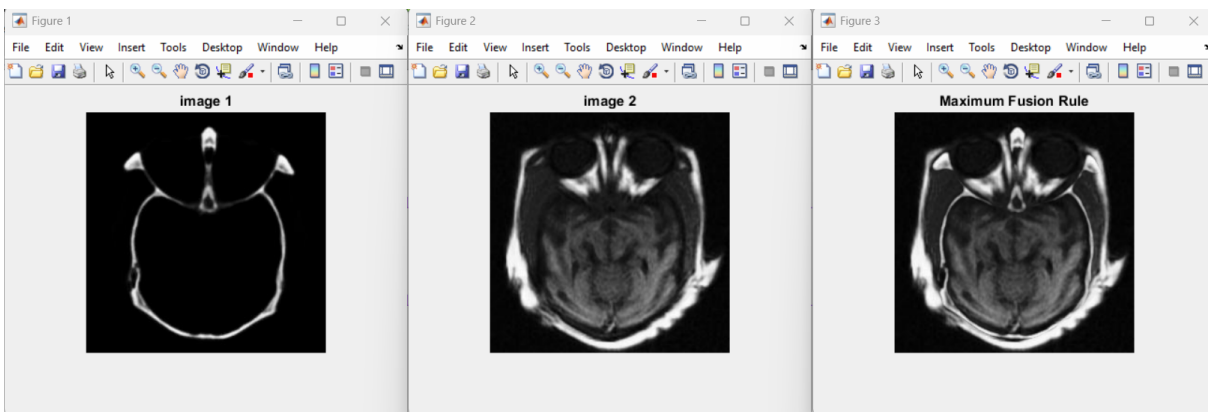

Figure 2
File Edit View Insert Tools Desktop Window Help
R

11.2 Image Fusion- Pixel level

```
%Image Fusion- Pixel level
 close all; clear all; clc;
I1=double(imread('C:\Program Files\MATLAB\MATLAB Production
Server\R2015a\bin\MIPIMAGES\med256A.jpg'));
if size(I1,3)==3
    I1=rgb2gray(I1);
end
I2=double(imread('C:\Program Files\MATLAB\MATLAB Production
Server\R2015a\bin\MIPIMAGES\med256B.jpg'));
if size(I2,3)==3
    I2=rgb2gray(I2);
end

for i=1:size(I1,1)
    for j=1:size(I1,2)
        I(i,j)= max(I1(i,j),I2(i,j));
    end
end

figure, imshow(I1,[]);
figure, imshow(I2,[]);
figure, imshow(I,[]);
title('Maximum Fusion Rule');
```



11.3 Image Fusion- Feature level

```
close all;
clear all;
clc;
I1=double(imread('C:\Program Files\MATLAB\MATLAB Production
Server\R2015a\bin\MIPIMAGES\med256A.jpg'));

if size(I1,3)==3
   I1=rgb2gray(I1);
end

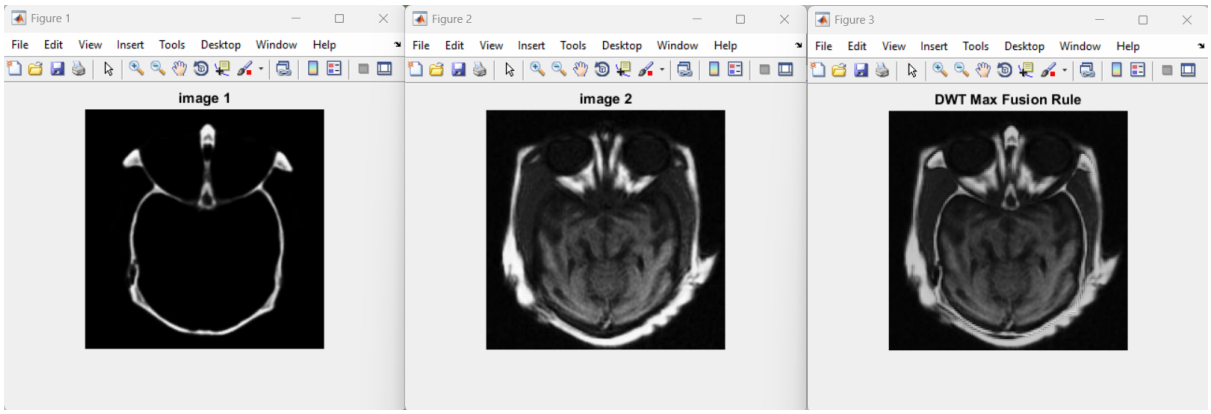
% source image 2

%  I2=double(imread('C:\Program Files\MATLAB\MATLAB Production
Server\R2015a\bin\MIPIMAGES\med256B.jpg'));
I2=double(imread('C:\Program Files\MATLAB\MATLAB Production
Server\R2015a\bin\MIPIMAGES\med256B.jpg'));

if size(I2,3)==3
   I2=rgb2gray(I2);
end
X=cell(1,4);
Y=cell(1,4);
Z=cell(1,4);
[X{1},X{2},X{3},X{4}] = dwt2(I1,'haar');
[Y{1},Y{2},Y{3},Y{4}] = dwt2(I2,'haar');

for i=1:4
   for j=1:size(X{1},1)
      for k=1:size(X{1},2)
         Z{i}(j,k)=max(X{i}(j,k),Y{i}(j,k));
      end
   end
```

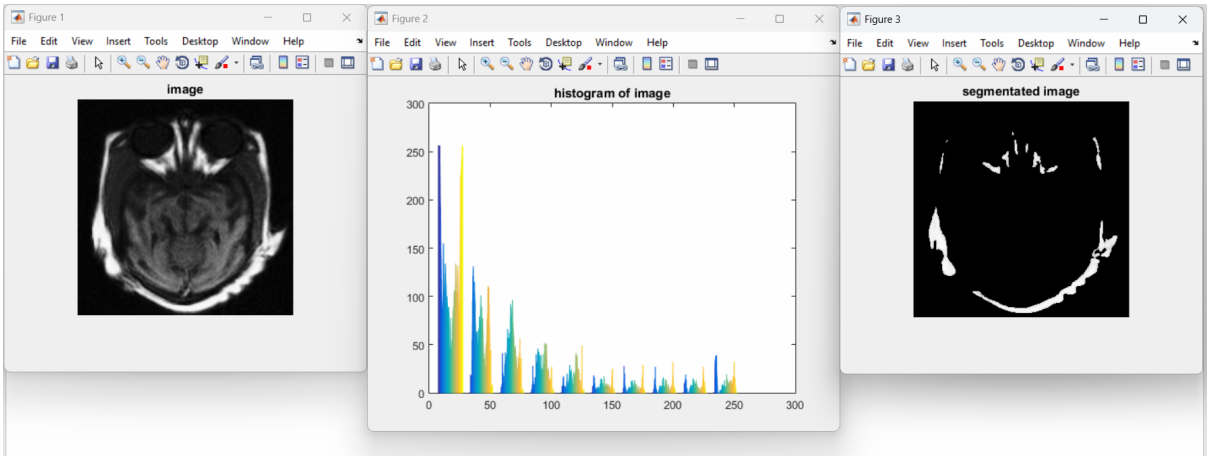
```
end
IF=idwt2(Z{1},Z{2},Z{3},Z{4},'haar');
figure, imshow(I1,[]);
title('image 1');
figure, imshow(I2,[]);
title('image 2');
figure, imshow(IF,[]);
title('DWT Max Fusion Rule');
```



12. Image Segmentation using Thresholding

```
% Image Segmentation using Thresholding
   close all; clear all; clc;
I1=double(imread('C:\Program Files\MATLAB\MATLAB Production
Server\R2015a\bin\MIPIMAGES\med256B.jpg'));
 if size(I1,3)==3
   I1=rgb2gray(I1);
end
figure(1);
imshow(I1,[]);
title('image');
figure(2);
hist(I1);
title('histogram of image');
[m n]=size(I1);

for i=1:m
   for j=1:n
      if I1(i,j) > 200
         I2(i,j)=I1(i,j);
      else
         I2(m,n)=0;
      end
   end
end
figure(3);
imshow(I2,[]);
title('segmentated image');
```


Figure 1
File Edit View Insert Tools Desktop Window Help
image
Figure 2
File Edit View Insert Tools Desktop Window Help
histogram of image
300
250
200
150
100
50
0
0 50 100 150 200 250 300
Figure 3
File Edit View Insert Tools Desktop Window Help
segmentated image